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Devices and methods for locating and implanting tissue anchors at mitral valve commissure

Abstract

A system is transluminally advanced into a subject. A first portion of the system is positioned adjacent a cardiovascular tissue of the subject, the system including wires disposed at the first portion. The first portion is stabilized at the cardiovascular tissue by positioning a distal portion of the system against a cardiovascular wall of the subject. While the distal portion of the system remains against the cardiovascular wall, the wires are induced to bow laterally such that the wires press against the cardiovascular tissue. While the wires remain pressed against the cardiovascular tissue, the shape of the wires is radiographically imaged. While (i) the first portion remains stabilized, and (ii) the wires remain pressed against the cardiovascular tissue, at least a portion of the system is transluminally implanted at the cardiovascular tissue, facilitated by the imaging. Other embodiments are also described.

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8460370	12/2012	Zakay	N/A	N/A

8460371	12/2012	Hlavka et al.	N/A	N/A
8475491	12/2012	Milo	N/A	N/A
8475525	12/2012	Maisano et al.	N/A	N/A
8480732	12/2012	Subramanian	N/A	N/A
8518107	12/2012	Tsukashima et al.	N/A	N/A
8523940	12/2012	Richardson et al.	N/A	N/A
8551161	12/2012	Dolan	N/A	N/A
8585755	12/2012	Chau et al.	N/A	N/A
8591576	12/2012	Hasenkam et al.	N/A	N/A
8608797	12/2012	Gross et al.	N/A	N/A
8628569	12/2013	Benichou et al.	N/A	N/A
8628571	12/2013	Hacohen et al.	N/A	N/A
8641727	12/2013	Starksen et al.	N/A	N/A
8652202	12/2013	Alon et al.	N/A	N/A
8652203	12/2013	Quadri et al.	N/A	N/A
8679174	12/2013	Ottma et al.	N/A	N/A
8685086	12/2013	Navia et al.	N/A	N/A
8728097	12/2013	Sugimoto et al.	N/A	N/A
8728155	12/2013	Montorfano et al.	N/A	N/A
8734467	12/2013	Miller et al.	N/A	N/A
8734699	12/2013	Heideman et al.	N/A	N/A
8740920	12/2013	Goldfarb et al.	N/A	N/A
8747463	12/2013	Fogarty et al.	N/A	N/A
8778021	12/2013	Cartledge	N/A	N/A
8784481	12/2013	Alkhatib et al.	N/A	N/A
8790367	12/2013	Nguyen et al.	N/A	N/A
8790394	12/2013	Miller et al.	N/A	N/A
8795298	12/2013	Hernlund et al.	N/A	N/A
8795355	12/2013	Alkhatib	N/A	N/A
8795356	12/2013	Quadri et al.	N/A	N/A
8795357	12/2013	Yohanan et al.	N/A	N/A
8808366	12/2013	Braido et al.	N/A	N/A
8808368	12/2013	Maisano et al.	N/A	N/A
8845717	12/2013	Khairkhahan et al.	N/A	N/A
8845723	12/2013	Spence et al.	N/A	N/A
8852261	12/2013	White	N/A	N/A
8852272	12/2013	Gross et al.	N/A	N/A
8858623	12/2013	Miller et al.	N/A	N/A
8864822	12/2013	Spence et al.	N/A	N/A
8870948	12/2013	Erzberger et al.	N/A	N/A
8870949	12/2013	Rowe	N/A	N/A
8888843	12/2013	Khairkhahan et al.	N/A	N/A
8889861	12/2013	Skead et al.	N/A	N/A
8894702	12/2013	Quadri et al.	N/A	N/A
8911461	12/2013	Traynor et al.	N/A	N/A
8911494	12/2013	Hammer et al.	N/A	N/A
8926696	12/2014	Cabiri et al.	N/A	N/A
8926697	12/2014	Gross et al.	N/A	N/A
8932343	12/2014	Alkhatib et al.	N/A	N/A
8932348	12/2014	Solem et al.	N/A	N/A

8940044	12/2014	Hammer et al.	N/A	N/A
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8951285	12/2014	Sugimoto et al.	N/A	N/A
8951286	12/2014	Sugimoto et al.	N/A	N/A
8961595	12/2014	Alkhatib	N/A	N/A
8961602	12/2014	Kovach et al.	N/A	N/A
8968335	12/2014	Robinson et al.	N/A	N/A
8979922	12/2014	Jayasinghe et al.	N/A	N/A
8992604	12/2014	Gross et al.	N/A	N/A
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9011530	12/2014	Reich et al.	N/A	N/A
9023100	12/2014	Quadri et al.	N/A	N/A
9072603	12/2014	Tuval et al.	N/A	N/A
9107749	12/2014	Bobo et al.	N/A	N/A
9119719	12/2014	Zipory et al.	N/A	N/A
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9125742	12/2014	Yoganathan et al.	N/A	N/A
9138316	12/2014	Bielefeld	N/A	N/A
9173646	12/2014	Fabro	N/A	N/A
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9180007	12/2014	Reich et al.	N/A	N/A
9192472	12/2014	Gross et al.	N/A	N/A
9198756	12/2014	Aklog et al.	N/A	N/A
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9265608	12/2015	Miller et al.	N/A	N/A
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9474606	12/2015	Zipory et al.	N/A	N/A
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9693865	12/2016	Gilmore et al.	N/A	N/A
9730793	12/2016	Reich et al.	N/A	N/A
9788941	12/2016	Hacohen	N/A	N/A
9801720	12/2016	Gilmore et al.	N/A	N/A
9907547	12/2017	Gilmore et al.	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a Continuation of U.S. patent application Ser. No. 17/173,883 to Alexander et al., filed Feb. 11, 2021; which is a Continuation of U.S. patent application Ser. No. 16/123,759 to Alexander et al., filed Sep. 6, 2018 (now U.S. Pat. No. 10,918,373); which is a Divisional of U.S. patent application Ser. No. 14/472,867, to Alexander et al., filed Aug. 29, 2014 (now U.S. Pat. No. 10,070,857), which is based on and claims priority to U.S. Provisional Patent Application 61/872,704 to Alexander et al., filed, Aug. 31, 2013, all of which are incorporated by reference, as if expressly set forth in their respective entireties herein.

TECHNICAL FIELD

(1) The present teachings generally relate to a delivery catheter system that is adapted to deliver multiple devices, such as guide wires, across the commissure of the mitral valve annulus. The present teachings further relate to implanting a first tissue anchor across the commissure of the mitral annulus. Other embodiments of the present teachings relate to pulling a ventricular free wall towards a septum.

BACKGROUND

(2) The left side of a human heart includes the left atrium (LA) and the left ventricle. The aorta receives oxygenated blood from the left ventricle through the aortic valve, which serves to prevent regurgitation of blood back into the left ventricle. The mitral valve is positioned between the left atrium and the left ventricle and allows a one-way flow of the oxygenated blood from the left atrium to the left ventricle.

(3) The mitral valve, which will be described below in greater detail, includes an anterior leaflet and a posterior leaflet that are coupled to the chordae tendineae. The commissures define an area where the anterior and posterior leaflets join and extend into the annulus. The chordae tendineae serve as “tension members” that prevent the leaflets of the mitral valve from moving past their dosing point and prolapsing back into the left atrium. When the left ventricle contracts during a systole, the chordae tendineae prevent the upward motion (toward the left atrium) of the anterior and posterior leaflets from passing the point where the anterior and posterior leaflets meet and seal to prevent backflow from the left ventricle to the left atrium (“mitral regurgitation” or “mitral insufficiency”). The chordae tendineae arise from the columnae carnea or, more specifically, the muscoli papillares (papillary muscles) of the columnae carnea. In various figures herein, some anatomical features have been deleted solely for clarity.

(4) The anterior leaflet and the posterior leaflet of the mitral valve are generally thin, flexible membranes. When the mitral valve is closed, the anterior leaflet and the posterior leaflet are generally aligned and contact each other along a “line of coaptation” several millimeters from their free edges to create a seal that prevents mitral regurgitation. Alternatively, when the mitral valve is opened, blood flows downwardly into the left ventricle through an opening created between the anterior leaflet and the posterior leaflet.

(5) Many problems relating to the mitral valve may cause many types of ailments. Such problems include, but are not limited to, mitral regurgitations. A mitral regurgitation, or leakage, is the backflow of blood from the left ventricle into the left atrium due to an imperfect closure of the mitral valve. Leakage often occurs when the anterior and posterior leaflets do not seal against each other, resulting in a mitral orifice between the anterior leaflet and the posterior leaflet when the leaflets are supposed to be fully coapted during a systole.

(6) In general, a relatively significant systolic mitral orifice may exist between the anterior leaflet and the posterior leaflet for a variety of reasons. For example, a mitral orifice may exist due to congenital malformations because of ischemic disease or the heart having been damaged by a previous heart attack. Such a mitral orifice may also be treated in a congestive heart failure, e.g., cardiomyopathy or some other type of distress which causes a heart to be enlarged. Enlargement of the heart can result in dilation (stretching) of the mitral annulus. This enlargement is usually limited to the posterior valve annulus and is associated with the posterior leaflet because the anterior annulus is a relatively rigid fibrous structure. When the posterior annulus enlarges, it causes the posterior leaflet to move away from the anterior leaflet, causing a mitral orifice during systoles because the two leaflets no longer form a proper coaptation. This results in leakage of blood through the valve or regurgitation.

(7) The blood leakage through the mitral valve generally causes a heart to operate less efficiently because the heart pumps blood both out to the body via the aorta and back (in the form of mitral regurgitation) into the left atrium. Leakage through the mitral valve, or general mitral insufficiency,

is thus often considered to be a precursor to congestive heart failures (CHF) or a cause to progressively worsening of a heart failure. There are generally different levels of symptoms associated with a heart failure. These levels are classified by the New York Heart Association (NYHA) functional classification system. The levels range from a Class 1 level which is associated with an asymptomatic patient who has substantially no physical limitations to a Class 4 level which is associated with a patient who is unable to carry out any physical activity without discomfort and has symptoms of cardiac insufficiency even at rest. In general, correcting or reducing the degree of mitral valve leakage can reduce a patient's NYHA grade. For instance, a patient with a Class 4 classification may have his classification reduced to Class 3 or Class 2 and, as a result, the patient becomes relatively comfortable at rest or even during a mild physical exertion. By eliminating the backflow of blood into the left atrium, therapies that reduce mitral insufficiency reduce the workload of the heart and may prevent or slow the degradation of heart function and congestive heart failure symptoms that are common when a significant degree of mitral insufficiency remains uncorrected.

(8) Treatments used to correct for mitral valve leakage or, more generally, CHF, are typically highly invasive, open-heart surgical procedures. In extreme cases, this may include implantation of a ventricular assist device such as an artificial heart in a patient with a failing heart. The implantation of a ventricular assist device is often expensive, and a patient with a ventricular assist device must be placed on an extended anti-coagulant therapy. An anti-coagulant therapy reduces the risk of blood clot formation, for example, within the ventricular assist device. Reducing the risks of blood clots associated with the ventricular assist device is desirable, but anti-coagulant therapies may increase the risk of uncontrollable bleeding in a patient, e.g., as a result of a fall.

(9) Rather than implanting a ventricular assist device, bi-ventricular pacing devices similar to pacemakers may be implanted in some cases, including those where a heart beats inefficiently in a particular asynchronous manner. While the implantation of a bi-ventricular pacing device may be effective, not all heart patients are suitable for receiving a bi-ventricular pacing device. Further, the implantation of a bi-ventricular pacing device is expensive, and is generally not effective in significantly reducing or eliminating the degree of mitral regurgitation/

(10) Open-heart surgical procedures that are intended to correct a mitral valve leakage, specifically, can involve the implantation of a replacement valve. Valves from animals, e.g., pigs, may be used to replace a mitral valve in a human. While a pig valve may relatively successfully replace a mitral valve, such replacement valves generally wear out, thereby requiring additional open surgeries at later dates. Mechanical valves, which are less likely to wear out may also be used to replace a leaking mitral valve. However, when a mechanical valve is implanted, there is an increased risk of thromboembolism and, as a result, the patient is generally required to undergo extended anti-coagulant therapies.

(11) A less invasive surgical procedure involves a heart bypass surgery through a port access procedure. For a port access procedure, the heart may be accessed by cutting between ribs or sometimes removing parts of one or more ribs, as opposed to dividing the sternum and opening the entire chest of a patient in an open-heart surgery.

(12) One open-heart surgical procedure that is particularly successful in correcting a mitral valve leakage and mitral regurgitation is an annuloplasty procedure. During an annuloplasty procedure, a medical device such as an annuloplasty ring may be implanted surgically on the left atrial side of the mitral annulus (i.e., the base of the mitral valve to the heart). The device reduces a dilated mitral valve annulus to a relatively normal size, moves the posterior leaflet closer to the anterior leaflet to aid anterior-posterior leaflet coaptation, and improves the quality of mitral valve closures during systoles. Annuloplasty rings are often shaped substantially like the letter "D" to correspond to the natural shape of the mitral annulus as viewed from above. Typically, the rings are formed from a rod or tube of a biocompatible material, including a plastic that has a DACRON mesh covering.

(13) In order for an annuloplasty ring to be implanted a surgeon surgically attaches the annuloplasty ring to the mitral valve on its atrial side. Conventional methods for installing a ring require an open-heart surgery which involves opening a patient's sternum and placing the patient on a heart-bypass machine. The annuloplasty ring is sewn on the top portion of the mitral valve. In sewing the annuloplasty ring onto the mitral valve, a surgeon generally sews the straight side of the "D" to the fibrous tissue located at the junction between the posterior wall of the aorta and the base of the anterior mitral valve leaflet. As the curved part of the ring is sewn to the posterior aspect of the annulus, the surgeon alternately acquires a relatively larger amount of tissue from the mitral annulus, e.g., a one-eighth inch bite of tissue, using a needle and thread, compared to a relatively smaller bite taken of the fabric covering of the annuloplasty ring. After the thread has loosely coupled the annuloplasty ring to the mitral valve annulus tissue, the annuloplasty ring is slid into contact with the mitral annulus. The tissue of the posterior mitral annulus that was previously stretched out, e.g., due to an enlarged heart, is effectively reduced in circumference and pulled forwards towards the anterior mitral leaflet by the tension applied by annuloplasty ring with the suture or thread. As a result, a mitral orifice between the anterior leaflet and the posterior leaflet may be reduced and even substantially closed off during a ventricular contraction or systole in many cases, thereby significantly reducing or even eliminating mitral insufficiency. After the mitral valve is shaped by the ring, the anterior and posterior leaflets will reform typically by pulling the posterior leaflet forward to properly meet the anterior leaflet and create a new contact line that will enable the mitral valve to function properly.

(14) Although a patient that receives an annuloplasty ring may be subjected to anti-coagulant therapies, the patient is only subjected to the therapies for a matter of weeks, e.g., until tissue grows over the annuloplasty ring.

(15) Another type of procedure that is generally effective in reducing mitral valve leakage associated with purpose of the valve leaflets involves placing a single edge-to-edge suture in the mitral valve. For example, in an Alfieri stitch or a bow-tie repair procedure, an edge-to-edge stitch is made approximately at the center of the mitral orifice between an anterior leaflet and a posterior leaflet of a mitral valve. Once the stitch is in place between the anterior and posterior leaflets, it is pulled in to form a suture which holds the anterior leaflet against the posterior leaflet.

(16) Another surgical procedure that reduces mitral valve leakage involves placing sutures along a mitral valve annulus around the posterior leaflet. These sutures may be formed as a double track, e.g., in two "rows" from a single strand of suture material. The sutures are tied off approximately at a central point of the posterior leaflet. Pledgets are often positioned under selected sutures to prevent the sutures from tearing through the annulus. When the sutures are tightened and tied on the circumference of the annulus may effectively be reduced to a desired size such that the size of a systolic mitral orifice between the posterior leaflet and the anterior leaflet may be reduced.

(17) While the invasive surgical procedures have proven to be effective in the treatment of mitral valve leakage, they often have significant drawbacks. Any time a patient undergoes an open heart surgery, there are risks of infection. Opening the sternum and using a cardiopulmonary bypass machine has also been shown to result in a significant incidence of both short and long term neurological deficits. Further, given the complexity of an open-heart surgery and the significant recovery time, people that are not greatly inconvenienced by CHF symptoms, e.g., people at a Class 1 classification, may choose not to have any corrective surgery. In addition, people that need an open heart surgery the most, e.g., people at a Class 4 classification, may either be too frail or too weak to undergo the surgery. Hence, many people that may benefit from a surgically repaired mitral valve may not undergo surgeries.

(18) Alternatively, in a minimally invasive method, by using a catheter system, a cinching device with distal, mid, and proximal anchors is placed within the lumen of the CS to allow plication of the annulus via the CS. In practice, these anchors are cinched together and the distance between them is shortened by pulling a flexible tensile member such as a cable or suture. As a result, the

valve annulus is shortened and the posterior leaflet is pulled closer to the anterior leaflet in a manner similar to an annuloplasty procedure. Unfortunately, since the tissue that forms the CS is relatively delicate, the anchors are prone to tear the tissue during the cinching procedure. In addition, the effect on the mitral annulus may be reduced when the CS of a particular patient is not directly aligned with the mitral annulus. Other minimally invasive techniques have been proposed but have various drawbacks related to such factors as effectiveness and/or accuracy of a catheter-based implementation.

(19) Catheter-based surgical procedures have been used to repair a defective mitral valve.

Specifically, anchors are secured at a plurality of locations around the annulus near the posterior leaflet of a mitral valve. Each anchor has a suture coupled thereto. The sutures are collectively gathered and pulled tight. As the sutures are pulled, the tissue between each pair of the adjacent anchors is plicated, thereby shortening the length of the annulus and drawing the posterior leaflet toward the anterior leaflet. Similar techniques can also be used to repair a defective tricuspid valve.

(20) During a surgical procedure, anchors are usually introduced and seemed sequentially. A typical repair by using the catheter based surgical procedure includes one or more steps selected from introducing a catheter to a proximity of the annulus, making an incision at the annulus, introducing a guide wire through the incision site, withdrawing the catheter, introducing an anchor by tracking a second catheter through the guide wire, securing the anchor in the annulus, and withdrawing the second catheter. This procedure is repeated to secure a second anchor.

(21) Catheters capable of delivering multiple guide wires or anchors have been disclosed. Without claiming to have exhaustively examined prior art references and without attempting to characterize any prior art reference, U.S. Patent Application Publication No. 2008-0228265 discloses a triple lumen catheter. However, distances between two of the three lumens are usually fixed. In addition, during a deployment, the two outer catheters are generally advanced lengthwise as well as laterally. In certain instances, one or both of the two outer catheters are caught by the chordae tendineae during a deployment.

(22) There generally exists a need for an improved catheter to simplify the catheter-based mitral valve correction.

SUMMARY

(23) One aspect of the present teachings provides a device for percutaneously locating the mitral valve commissure. In various embodiments, the device comprises a bow catheter, a central catheter, and two bow wires. In various embodiments, the bow catheter has a proximal end, a distal end, and an elongated tubular body. In various embodiments, the central catheter also has a proximal end, a distal end, and an elongated tubular body, wherein the central catheter slidably is disposed within a lumen of the bow catheter. And in some embodiments, two bow wires each has a proximal end attaching to the distal end of the bow catheter and a distal end attaching to the distal end of the central catheter. In some embodiments, as the distal ends of the bow catheter and central catheter come closer to each other, the two bow wires bend radially outward, forming a general plane.

(24) Another aspect of the present teachings provides a device for percutaneously implanting a wire across the mitral annulus at the mitral valve commissure. In various embodiments, this device comprises a bow catheter, a central catheter, two bow wires, and at least one wire delivery catheter. In some embodiments, the bow catheter has a proximal end, a distal end, and an elongated tubular body. In some embodiments, the central catheter also has a proximal end, a distal end, and an elongated tubular body, wherein the central catheter slidably is disposed within a lumen of the bow catheter. In some embodiments, the two bow wires each has a proximal end attaching to the distal end of the bow catheter and a distal end attaching to the distal end of the central catheter. In some embodiments, as the distal ends of the bow catheter and central catheter come closer to each other, the two bow wires bend radially outward forming a general plane. In some embodiments, the at least one wire delivery catheter has a distal portion attaching to a portion of one of the two bow wire. In some embodiments, as the two bow wires bend radially outward, the distal portion of the

wire delivery catheter is directed to pivot outward radially from an elongated axis of the wire delivery catheter.

(25) Another aspect of the present teachings provides a method for percutaneously locating a mitral valve commissure. In various embodiments, the method comprises providing a mitral valve commissure locating system. In some embodiments, the mitral valve commissure locating system comprises a bow catheter having a proximal end, a distal end, and an elongated tubular body; a central catheter having a proximal end, a distal end, and an elongated tubular body; and two bow wires with proximal ends and distal ends. In some embodiments, the central catheter slidably is disposed within a lumen of the bow catheter. In some embodiments, the proximal ends of the bow wire attach to the distal end of the bow catheter and the distal ends of the bow wire attach to the distal end of the central catheter. And in some embodiments, as the distal ends of the bow catheter and central catheter come closer to each other, the two bow wires bend radially outward to form a general plane. In various embodiments, the method also comprises delivering the mitral valve commissure locating system percutaneously across the mitral valve with the distal ends of the bow wires distal to the mitral annulus and the proximal ends of the bow wires proximal to the mitral annulus. In various embodiments, the method comprises rendering the distal end of the bow catheter and the distal end of the central catheter closer to each other and the two bow wires bending radically outward. In some embodiments, the method also comprises visualizing the curvature of the bow wires as it keeps bending radially outward and discontinuing outward bending of the bow wires when a narrower waist forms along the curvature of the bow wires.

(26) Another aspect of the present teachings provides a method for percutaneously implanting a wire across the mitral annulus at the mitral valve commissure. In various embodiments, the method comprises providing a mitral valve commissure locating system and at least one wire delivery catheter attaching to the mitral valve commissure locating system. In various embodiments, the mitral valve commissure locating system comprises a bow catheter comprising a proximal end, a distal end, and an elongated tubular body; a central catheter comprising a proximal end, a distal end, and an elongated tubular body; and two bow wires each comprising a proximal end and a distal end. In some embodiments, the central catheter slidably is disposed within a lumen of the bow catheter. In some embodiments, the proximal ends of the bow wire attach to the distal end of the bow catheter and the distal ends of the bow wire attach to the distal end of the central catheter. And as the distal ends of the bow catheter and central catheter come closer to each other, the two bow wires bend radially outward and form a general plane. In certain embodiments, the at least one wire delivery catheter comprises a distal portion attaching to a portion of one of the two bow wire. In certain embodiments, as the two bow wires bend radially outward, the distal portion of the wire delivery catheter is directed to pivot outward radially from an elongated axis of the wire delivery catheter. In various embodiments, the method comprises delivering the mitral valve commissure locating system percutaneously across the mitral valve, with the distal ends of the bow wires distal to the mitral annulus and the proximal ends of the bow wires proximal to the mitral annulus. In some embodiments, the method comprises rendering the distal end of the bow catheter and the distal end of the central catheter closer to each other and the two bow wires bending radially outward. In some embodiments, the method comprises visualizing the curvature of the bow wires as they keep bending radially outward, and discontinuing outward bending of the bow wires when a narrower waist forms along the curvature of the bow wires. In some embodiments, the method comprises positioning a distal end of the wire delivery catheter proximally and adjacent to the mitral annulus, and advancing a wire distally through the lumen of the wire delivery catheter across the mitral annulus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. **1a-1b** illustrate the anatomy of a left heart.
- (2) FIGS. **2a-2b** are perspective views of an exemplary mitral commissure locating system in accordance with the present teachings.
- (3) FIGS. **3a-3b** are perspective views of an exemplary mitral commissure locating system percutaneously inserted across the mitral valve in accordance with certain embodiments of the present teachings.
- (4) FIG. **3c** is a perspective view of another exemplary mitral commissure locating system in accordance with the present teachings.
- (5) FIG. **3d** is a perspective view of another exemplary mitral commissure locating system percutaneously inserted into the left heart in accordance with certain embodiments of the present teachings.
- (6) FIGS. **4a-4b** are perspective views of an exemplary tissue piercing wire percutaneously deployed across the mitral annulus at a commissure in accordance with certain embodiments of the present teachings.
- (7) FIGS. **5a-5b** are perspective views of an exemplary mitral commissure locating system in accordance the present teachings.
- (8) FIG. **6** is a perspective view of two exemplary tissue piercing wire percutaneously deployed across the mitral annulus at both the commissure in accordance with certain embodiments of the present teachings.
- (9) FIG. **7** is a perspective view of an exemplary tissue anchor delivery catheter percutaneously tracking over an exemplary tissue piercing wire across mitral annulus at a commissure in accordance with certain embodiments of the present teachings.
- (10) FIG. **8** is a perspective view of an exemplary tissue anchor in accordance with the present teachings.
- (11) FIGS. **9a-9c** are perspective views of an exemplary tissue anchor percutaneously deployed across the mitral annulus at a commissure in accordance with certain embodiments of the present teachings.
- (12) FIG. **10** is a perspective view of two exemplary tissue anchors percutaneously deployed across the mitral annulus at both commissures in accordance with certain embodiments of the present teachings.
- (13) FIGS. **11a-11d** are perspective views of multiple exemplary tissue anchors percutaneously deployed across the mitral annulus in accordance with certain embodiments of the present teachings.
- (14) FIGS. **12a-12b** are perspective views of tissue plication with multiple exemplary tissue anchors across the mitral annulus in accordance with certain embodiments of the present teachings.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

(15) Certain specific details are set forth in the following description and Figures to provide an understanding of various embodiments of the present teachings. Those of ordinary skill in the relevant art will understand that they can practice other embodiments of the present teachings without one or more of the details described herein. Thus, it is not the intention of the Applicants to restrict or in any way limit the scope of the appended claims to such details. While various processes are described with reference to steps and sequences in the following disclosure, the steps and sequences of steps should not be taken as required to practice all embodiments of the present teachings. Thus, it is not the intention of the Applicants to restrict or in any way limit the scope of the appended claims to such steps or sequences of steps.

(16) As used herein, the terms “subject” and “patient” refer to an animal, such as a mammal, including livestock, pets, and preferably a human. Specific examples of “subjects” and “patients” include, but are not limited to, individuals requiring medical assistance and, in particular, requiring

treatment for symptoms of a heart failure.

(17) As used herein, the term “lumen” means a canal, duct, generally tubular space or cavity in the body of a subject including veins, arteries, blood vessels, capillaries, intestines, and the like. The term “lumen” can also refer to a tubular space in a catheter, a sheath, or the like in a device.

(18) As used herein, the term “proximal” means close to the operator (less into the body) and “distal” shall mean away from the operator (further into the body). In positioning a medical device from a downstream access point, distal is more upstream and proximal is more downstream.

(19) As used herein, the term “catheter” or “sheath” encompasses any hollow instrument capable of penetrating a body tissue or interstitial cavities and providing a conduit for selectively injecting a solution or gas. The term “catheter” or “sheath” is also intended to encompass any elongate body capable of serving as a conduit for one or more of the ablation, expandable or sensing elements. Specifically, in the context of coaxial instruments, the term “catheter” or “sheath” can encompass either the outer catheter body or sheath or other instruments that can be introduced through such a sheath. The use of the term “catheter” should not be construed as meaning only a single instrument but rather is used to encompass both singular and plural instruments, including coaxial, nested, and other tandem arrangements. Moreover, the terms “sheath” or “catheter” are sometime used interchangeably to describe catheters having at least one lumen through which instruments or treatment modalities can pass.

(20) Unless otherwise specified, all numbers expressing quantities, measurements, and other properties or parameters used in the specification and claims are to be understood as being modified in all instances by the term “about”. Accordingly, unless otherwise indicated, it should be understood that the numerical parameters set forth in the following specification and attached claims are approximations. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, numerical parameters should be read in light of the number of reported significant digits and the application of ordinary rounding techniques.

(21) An aspect of the present teachings provides a positioning catheter system deploys across the mitral orifice between the anterior and posterior leaflets, making out two mitral valve commissures. In various embodiments, the positioning catheter system includes a central catheter, a guide wire, and a bow catheter with two bow wires. In various embodiments upon deployment, at least one of the two bow wires bends radially outward, along the slit opening of the mitral orifice until it reaches the commissure.

(22) Another aspect of the present teachings provides a delivery catheter system for delivering multiple wires across the mitral annulus at the mitral valve commissure. In various embodiments, the delivery catheter system comprises a positioning catheter system. In some embodiments, a wire delivery catheter tracks along the positioning catheter system and deploys a wire across the mitral annulus at the mitral valve commissure, in other embodiments, the delivery catheter system includes two wire delivery catheters each tracking along the bow wire and reaching the mitral valve commissure. In some embodiments, each wire delivery catheter independently deploys one wire across the mitral annulus at the mitral valve commissure.

(23) Another aspect of the present teachings provides a method of plicating mitral annulus tissue around the mitral valve commissure. Upon deploying wire across the mitral annulus at the mitral valve commissure, another aspect of the present teachings further discloses implanting two tissue anchors across the annulus at the mitral valve commissure with a pre-defined distance in between. The present teachings further disclose reducing the distance between the two tissue anchors, thereby plicating the annulus tissues.

(24) The following description refers to FIGS. 1 to 12. A person with ordinary skill in the art would understand that the figures and description thereto refer to various embodiments of the present teachings and, unless indicated otherwise by their contexts, do not limit the scope of the attached claims.

(25) FIGS. 1a-b illustrate a mitral valve (6) between the left atrium (2) and the left ventricle (4) of

the heart. Mitral valve (6) includes an anterior leaflet (8) and a posterior leaflet (10) that converge at the posteromedial commissure (12) and the anterolateral commissures (14). The semi-circular anterior leaflet (8) is anchored anteriorly to one third of the mitral annulus (16a) and the quadrangular posterior leaflet (10) is attached posteriorly to the remaining two thirds of the posterior mitral annulus (16b). The posterior leaflet (10) has two indentations, dividing the posterior leaflet into a lateral P1 scallop close to the anterolateral commissure (14), a larger central P2 scallop, and a medial P3 scallop close to the posteromedial commissure (12). Arbitrarily opposing segments of the anterior leaflet (8) are called A1; A2, and A3 scallops, respectively.

(26) The normal anterior and posterior leaflets (8, 10) close (coapt) from their lateral to medial portions (coaptation line) during a systole, preventing blood from back flowing into the left atrium (2). The opening between the anterior and posterior leaflets (8, 10) is surrounded by a fibrous ring known as the mitral annulus (16), which resides in the left atrio-ventricular groove. The mitral annulus (16) consists of an anterior part (16a) and a posterior part (16b).

(27) There are two papillary muscles (18, 20) arising from the area between the apical and middle thirds of the left ventricular wall, the anterolateral papillary muscles (18) and the posteromedial papillary muscle (20). Several dozens of tendinous chords (22) originate from these papillary muscles (18, 20) and attach to the ipsilateral half of the anterior and posterior mitral leaflets (8, 10).

(28) FIGS. 2a-b illustrate an embodiment of the mitral commissure locating system (50). According to one embodiment of the present teachings, the mitral commissure locating system (50) comprises a central catheter (60), a bow catheter (70), a guide wire (80), and two bow wires (30, 40). Each of the bow catheter (70) and the central catheter (60) has a proximal end (not shown), a distal end (62, 72), and an elongated tubular body (64, 74) comprising an axial lumen (66, 76) extending between the proximal and distal ends (62, 72). The guide wire (80) also comprises a proximal end (not shown), a distal end (82), and an elongated wire body (84) extending between the proximal (not shown) and distal ends (82). The proximal ends of the bow catheter (70), the central catheter (60) and the guide wire (80) are manipulated by a clinician from outside of the body. The distal ends (62, 72, 82) of the central catheter (60), the bow catheter (70), and the guide wire (80) are inserted into the heart percutaneously.

(29) The guide wire (80) is slidably disposed within the axial lumen (66) of the central catheter (60). A clinician can extend the guide wire (80) distally so that the distal end (82) of the guide wire (80) is outside of the distal end (62) of the central catheter (60). A clinician can also retract the guide wire (80) proximally so that the distal end (82) of the guide wire (80) is within the axial lumen (66) of the central catheter (60).

(30) The central catheter (60) is slidably disposed within the axial lumen (76) of the bow catheter (70). The bow wires (30, 40) also have proximal ends (34, 44), distal ends (32, 42), and elongated bow wires (36, 46) between the two ends. As illustrated in FIG. 2a, the distal ends (32, 42) of the bow wires (30, 40) are fixed to the central catheter (60) at or near its distal end (62), the proximal ends (34, 44) of the bow wires (30, 40) are fixed to the bow catheter (70) at or near its distal end (72). As the clinician extends the central catheter (60) distally, the distal end (62) of the central catheter (60) extends distally outside of the distal end (72) of the bow catheter (70). The length of bow wires (30, 40) limit the extent of the distal end (62) of the central catheter (60) beyond the distal end (72) of the bow catheter (70), as shown in FIG. 2a. As the clinician retracts the central catheter (60) proximally, the distal end (62) of the central catheter (60) moves proximally toward the distal end (72) of the bow catheter (70). As the proximal (34, 44) and distal ends (32, 42) of the bow wires (30, 40) draw closer, elongated bow wires (36, 46) extend radially outward as illustrated in FIG. 2b.

(31) According to one embodiment of the present teachings, the mitral commissure locating system (50) comprises an elongated delivery profile and an expanded deployed profile. FIG. 2a illustrates an elongated delivery profile of the mitral commissure locating system (50). In this delivery profile, the central catheter (60) extends distally from within the axial lumen (76) of the bow catheter (70).

The bow wires (30, 40) are in a generally straightened shape. The distal end (62) of the central catheter (60) and the distal end (72) of the bow catheter (70) are far apart from each other along the length of the bow wires (30, 40). The bow catheter (70), the bow wires (30, 40), and the central catheter (60) are packed tightly in their smallest radial profiles to be delivered through a delivery sheath (not shown).

(32) FIG. 2b illustrates a deployed profile of the mitral commissure locating system (50). In this deployed profile, the central catheter (60) retracts proximally, drawing its distal end (62) closer to the distal end (72) of the bow catheter (70), the bow wires (30, 40) arc outward radially away from the longitudinal axis of the bow catheter (70). The portion of the central catheter (60) outside of the bow catheter (70) and the radically curved bow wires form a generally flat surface that is configured to fit at the slit between the posterior (10) and anterior mitral leaflets (8). Alternatively, the radially curved bow wires (30, 40) form a gradually curved surface that fit within the curvature of the coaptation line from one commissure, across the mid anterior leaflet, to the other commissure so that valve leaflets are prevented from being held open during subsequent procedure.

(33) In some embodiments of the present teachings, the bow wires (30, 40) are made of a material conventionally used for guide wire (80). Examples of the material include a straight stainless steel wire, a coiled stainless steel wire, a glass fiber, a plastics material, nitinol, and etc. In one embodiment, the bow wires (30, 40) are made of an adaptable material that is configured to curve radically outward as its distal (32, 42) and proximal ends (34, 44) come together. In another embodiment, the bow wires (30, 40) are made of an adaptable material that at least a portion of the bow wires (30, 40) arc radially as the bow wires (30, 40) encounters tissues of a heart, for example, as described later in the present teachings.

(34) According to one embodiment of the present teachings, the mitral commissure locating system (50) transitions from its delivery profile to its deployed profile by a clinician from outside of the body. That is, a clinician retracts the proximal end of the central catheter (60) proximally while holding the bow catheter (70) steady, reduces the distance between the distal ends of the central catheter (60) and bow catheter, and expands the bow wire radially. Alternatively, a clinician can push the proximal end of the bow catheter (70) distally while holding the central catheter (60) steady, thereby deploying the mitral commissure locating system (50). One skilled in the art should understand that mechanisms, steps, and details of the deployment can vary. What has been disclosed in the present teachings is only exemplary and should not be viewed as limiting.

(35) Now referring to FIGS. 3a-b, a mitral commissure locating system (50) is delivered and deployed at a mitral valve (6) according to one embodiment of the present teachings. What is described below are certain exemplary mitral commissure locating system (50) of the present teachings. One ordinarily skilled in the art would understand that other ways of percutaneous delivery can also be used without departing from the spirit of the present teachings. Thus the disclosure below should not be viewed limiting.

(36) According to one embodiment of the present teachings, a delivery sheath (48) is directed into the aorta (24), through the aortic valve and into the left ventricle (4) between the tendinous chords (22). This delivery sheath (48) is then used as a conduit for the subsequent device delivery to the treatment site. Thus, the mitral commissure locating system (50) can be advanced through the delivery sheath (48) to the treatment location. One skilled in the art should understand, a delivery sheath (48) may not be necessary and the mitral commissure locating system (50) can be directly advanced to the treatment location.

(37) FIG. 3a illustrates an exemplary mitral commissure locating system (50) being delivered across the mitral valve (6). According to certain embodiments of the present teachings, the mitral commissure locating system (50) in its delivery profile is delivered to the mitral valve (6). The mitral commissure locating system (50) comprises a distal portion (52). As illustrated in FIG. 3a, part of the distal portion (52) of the mitral commissure locating system (50), for example the portions of the system comprising the distal half of the bow wires (30, 40), is disposed inside of the

left atrium (2) and the rest portion of the mitral commissure locating system (50) is disposed inside the left ventricle (4).

(38) In one embodiment of the present teachings, the guide wire (80) is first inserted through the delivery sheath (48) and enters the left ventricle (4). A clinician can manipulate the guide wire (80) so that the distal end (82) of the guide wire (80) is steered to enter between the papillary muscles (18, 20), through the slit between the posterior (10) and anterior mitral leaflets (8), and into the left ventricle (4). In some embodiments, the distal end (82) of the guide wire (80) can be turned, rotated, or deflected. The delectability or steerability of the guide wire (80) allows a clinician to manipulate the distal end (82) of the guide wire (80) from outside of the body and advance to the mitral annular slit between the posterior (10) and anterior mitral leaflets (8), and further distally to the left pulmonary vein when needed. Design and construction of a steerable and deflectable guide wire (80) are known to those with ordinary skill in the art.

(39) The central catheter (60), along with the bow catheter (70) and bow wires (30, 40), then tracks over the guide wire (80), so that the distal portion (52) of the mitral commissure locating system (50) is positioned across the mitral valve (6) as shown in FIG. 3a. Alternatively, the entire distal portion (52) of the mitral commissure locating system (50) including the guide wire (80), the central catheter (60), the bow wires (30, 40) and the bow catheter (70), is delivered together through the delivery sheath (48) and positioned across the mitral valve (6).

(40) FIG. 3b illustrates the above described exemplary mitral commissure locating system (50) being deployed across the mitral valve (6). As the distal ends (62, 72) of the central catheter (60) and bow catheter (70) draw closer by a clinician from outside of the body, the bow wires (30, 40) curves outward. As the distal portion (52) of the mitral commissure locating system (50) is deployed, the generally flat plane formed by the bow wires (30, 40) automatically aligns along the slit between the posterior (10) and anterior mitral leaflets (8). As the bow wires (30, 40) further axes outward, each wire makes contact with a mitral valve commissure (12, 14). As the distal ends (62, 72) of the central catheter (60) and bow catheter (70) is drawn further closely by a clinician from outside of the body, the bow wires (30, 40) curves further outward, portions of the bow wires (30, 40) contact the mitral valve commissure (12, 14) and form narrow waists (38, 48) as they are obstructed from expanding further radially, as illustrated in FIG. 3b.

(41) In an alternative embodiment, the distal portion of the mitral commissure locating system could be positioned inside a left pulmonary vein in order to stabilize the system for subsequent procedures. To achieve this, according to one embodiment of the present teachings, the bow wires are made long enough so that a proximal portion of the bow wires remain inside the left ventricle, a middle portion of the bow wires are inside the left atrium, and a distal portion of the bow wires are inside the left pulmonary vein. When deployed, as the bow wires curve radially outward, the radially expanded distal portion of the bow wire secures the distal position of the mitral commissure locating system inside the left pulmonary vein, and the radially expanded middle and proximal portions of the bow wires securely locked into the commissure as the heart contracts,

(42) FIG. 3c illustrates another embodiment of the present teaching, where the distal portion (302) of the mitral commissure locating system (300) has a second bow catheter-bow wire system which allows the distal portion (302) of the mitral commissure locating system (300) to be deployed inside the left pulmonary vein (28), thereby by stabilizing the positioning of the mitral commissure locating system (300). One embodiment of such mitral commissure locating system (300) includes a central catheter (310) attached to a second pair of the bow wires (330a, 330b) at its distal end (312). The distal ends (332a, 332b) of the second pair of the bow wires (330a, 330b) attach to the distal end of the central catheter (310). The proximal ends (334a, 334b) of the second pair of the bow wires (330a, 330b) attach to the distal end (342) of the bow catheter (340). Similar to what has been described above, the central catheter (310) is slidably disposed within the longitudinal lumen (344) of the bow catheter (340). The bow catheter (340) further comprises a pair of side openings (350a, 350b) along its tubular wall. A first pair of bow wires (320a, 320b) have their proximal ends

(324a, 324b) attaching to the proximal ends (354a, 354b) of the side openings (350a, 350b) on the bow catheter (340), and their distal ends (322a, 322b) attaching to the central catheter (310) along its tubular wall. The side openings (350a, 350b) are configured to allow the first pair of the bow wires (320a, 320b) to bend radially outward so that it can be locked inside the commissures (12, 14) when the distal end (312) of the bow catheter (340) and the distal end (342) of the central catheter (310) are drawn closer. The distance between the distal ends (322a, 322b) of the first pair of the bow wires (320a, 320b) and the proximal ends (334a, 334b) of the second pair of the bow wires (330a, 330b) are configured in such way that as the second pair of the bow wires (330a, 330b) are positioned inside the left pulmonary vein (28), the first pair of the bow wires (320a, 320b) are positioned across the mitral valve (6) between the mitral leaflets (8, 10). As the distal end (312) of the central catheter (310) is drawn closer to the distal end (342) of the bow catheter (340), the distal ends (322a, 322b, 332a, 332b) and proximal ends (324a, 324b, 334a, 334b) of both pairs of the bow wires (320a, 320b, 330a, 330b) are drawn closer at the same time. Both pairs of the bow wires (320a, 320b, 330a, 330b) curve radially outwardly. The first pair of bow wires (320a, 320b) are locked inside the commissures (12, 14) as the heart contracts, and the second pair of the bow wires (330a, 330b) secure the distal portion (302) of the mitral commissure locating system (300) inside the left pulmonary vein (28) as illustrated in FIG. 3d. One skilled in the art should understand that other mechanism which allows the distal portion of the mitral commissure locating system to be secured inside the left pulmonary vein can also be incorporated with the embodiment disclosed herein. For example, two central catheter-bow wire systems are constructed and put together, one to be deployed inside the left pulmonary vein, and the other to be deployed across the mitral valve. Thus, the specific embodiment provided in the present teachings should not be viewed as limiting. In one embodiment of the present teachings, the bow wires can include a radiopaque or echogenic marker so that the wire can be easily visualized by using radiographic imaging equipment such as with x-ray, magnetic resonance, ultrasound, or fluoroscopic techniques. The radiopaque marker can be in the form of threads, beads, or other forms. In some embodiments, radiopaque markers are attached to the wire. For example, the radiopaque markers can be wrapped, laminated, and/or bonded through a welding process. In another embodiment, the bow wire could be made of radiopaque material.

(43) In one embodiment, by visualizing the narrow waist (38, 48) on the bow wires (30, 40), a clinician identifies the mitral valve commissure (12, 14). In another embodiment, by forming the narrow waist (38, 48) on the bow wires (30, 40), the deployed distal portion (52) of the mitral commissure locating system (50) is securely positioned across the mitral valve (6) for subsequent procedures.

(44) In another embodiment, the bow wires of the mitral commissure locating system comprises a preformed reversed bends on the bow wires, which forms a narrow waist when deployed. In some embodiments, such reversed bends or the narrow waist fit into the mitral commissure and, as a result secure the mitral commissure locating system in the mitral annulus. Without being limited to any theory, such reversed bends/narrow waist of the mitral commissure locating system ensures a better positioning of the delivery catheter for the tissue piercing wire so that the distal end of the tissue piercing wire can be positioned close to the annulus.

(45) In one embodiment of the present teachings, upon marking the mitral valve commissure (12, 14) by the mitral commissure locating system (50), a clinician can then advance a delivery catheter for tissue piercing wire (device) (90) distally and position it adjacent, approximate to/or against the mitral annulus (16) at the mitral valve posteromedial commissure (12) from inside the left ventricle (4), as illustrated in FIG. 4a. In various embodiments of the present teachings, a tissue piercing wire (100) is pre-loaded within the lumen (96) of a delivery catheter for tissue piercing wire (90) during advancement of the delivery catheter of the tissue piercing wire (90) to the treatment site. In various other embodiments, the tissue piercing wire (100) is advanced through the lumen (96) after the delivery catheter of the tissue piercing wire (90) is placed at the treatment location.

(46) Once a delivery catheter is properly positioned, a tissue piercing wire (100) is advanced relative to the delivery catheter for tissue piercing wire (90) as illustrated in FIG. 4b. According to one embodiment, the distal end (102) of the tissue piercing wire (100) has a piercing tip which allows it to perforate the mitral annulus (16). In another embodiment, a radio frequency energy is then applied so that the distal piercing tip of the tissue piercing wire (100) is advanced through the mitral annulus (16) and reaches the left atrium (2). In yet another embodiment, the distal end of the delivery catheter is configured to pierce through the annulus. A wire then slides through the lumen of the delivery catheter to reach the opposite side of the annulus. One skilled in the art should understand that other structures that can serve the function of the tissue piercing wire are also included in the present teachings. Thus, the above discussion on tissue piercing wires should not be considered as limiting.

(47) In some embodiments, the movement of a tissue piercing wire (100) is accomplished manually. Alternatively, the movement of a tissue piercing wire (100) may be automated and therefore requires additional controls such as a spring-loaded mechanism attached to the delivery.

(48) FIGS. 5a-5b illustrate another embodiment of the present teachings. In one embodiment, one delivery catheter for the tissue piercing wire (110, 120) attaches to each one of the bow wire (30, 40) of the mitral valve commissure system (50). Each delivery catheter for tissue piercing wire (110, 120) includes a proximal end (not shown) being manipulated by a clinician from outside of the body, a distal end (112, 122), and a tubular elongated body (114, 124) with an axial lumen (116, 126). According to some embodiments, the distal ends (112, 122) of the delivery catheter for tissue piercing wire (110, 120) are distal to the distal end (62) of the bow catheter (70) by a pre-defined distance. As illustrated in FIGS. 5a-5b, in one embodiment of the present teachings, a distal portion (118, 128) of each delivery catheter for tissue piercing wire (110, 120) attaches to a portion of the bow wire (30, 40) and the rest portion of the tissue piercing catheter (110, 120) connects to a portion of the bow catheter (70). These attachments allow the distal portion (118, 128) of each delivery catheter for tissue piercing wire (110, 120) to move along with the movement of each of the bow wires (30, 40). That is, as the distal ends (62, 72) of the central catheter (60) and the bow catheter (70) come together, the bow wires (30, 40) are radially outwardly and the distal portion (118, 128) of delivery catheter for tissue piercing wires (110, 120) are directed radially outwardly by the bow wires (30, 40) as illustrated in FIG. 5b.

(49) In one embodiment, only portions of the delivery catheter for tissue piercing wire (110, 120) attaches to the bow catheter (70) and the bow wires (30, 40). In another embodiment, the entire length of the delivery catheter for tissue piercing wire (110, 120) either attaches to the bow wires (30, 40) or to the bow catheter (70). One skilled in the art should understand that the attachment between the delivery catheter for tissue piercing wire (110, 120) and the bow wires (30, 40) and the bow catheter (70) could vary according to the need of the design and function that needs to be achieved. For example, as the exemplary embodiment shown in FIG. 5b, a gap is present between the attachments of the delivery catheter for tissue piercing wire (110, 120) to the bow wires (30, 40) and to the bow catheter (70), which facilitates the radial bend of the distal portion (118, 128) of the delivery catheter for tissue piercing wire (110, 120) as shown. In another exemplary embodiment, also shown in FIG. 5a, the very distal end portion of the delivery catheter for tissue piercing wire (110, 120) is not attached to the bow wires (30, 40), so that as the bow wires (30, 40) is deployed and marks the mitral valve commissure (12, 14), the distal end portion of the delivery catheter for tissue piercing wire (110, 120) directs further radially outward at a pre-defined angle and distance so that the distal ends (112, 122) of the delivery catheter for tissue piercing wire (110, 120) can be positioned precisely at the most desired treatment location.

(50) According to one embodiment, the delivery catheter for tissue piercing wire (110, 120) attaches to the bow wires (30, 40) and the bow catheter (70) by a mechanical means including screws, bolts, clamps, bands, wire wraps, metal forms, or the like. According to another embodiment, the delivery catheter for tissue piercing wire (110, 120) attach to the bow wires (30,

40) and the bow catheter (**70**) by a chemical means, including an adhesive or the like. According to another embodiment, the delivery catheter for tissue piercing wire (**110, 120**) attach to the bow wires (**30, 40**) and the bow catheter (**70**) by chemical means, including ultrasonic welding, laser welding, overmolding, or the like. According to another embodiment, the delivery catheter for tissue piercing wire (**110, 120**) attach to the bow wires (**30, 40**) and the bow catheter (**70**) by other attachment means known to those skilled in the art. Yet in another embodiment, the delivery catheter for tissue piercing wire and the bow catheter are a single continuous tubular construction with an exit port cut on the tubular wall. In another embodiment, the tissue piercing wire is slidably disposed within the single lumen and exits the exit port on the tubular wall.

(51) In one embodiment, the delivery catheter for tissue piercing wires (**110, 120**) are stationarily attached to the bow catheter (**70**) and bow wires (**30, 40**). That is, the delivery catheter for tissue piercing wires (**110, 120**) cannot move relative to the bow catheter (**70**) and bow wires (**30, 40**). In another embodiment, the delivery catheter fix tissue piercing wires (**110, 120**) are slidably attached to the bow catheter (**70**) and bow wires (**30, 40**). That is, the delivery catheter for tissue piercing wires (**110, 120**) could slide against the bow catheter (**70**) and bow wires (**30, 40**) while still remaining attached to the bow wires (**30, 40**) and bow catheter (**70**),

(52) In one embodiment, one delivery catheter for tissue piercing wire attaches to each of the bow wires. In another embodiment, to achieve a small delivery profile, only one delivery catheter for tissue piercing wire attaches to only one of the bow wires. Thus, the specific disclosure herein should not be viewed as limiting.

(53) Similar to what has been described above, this embodiment of the present teachings also has an elongated delivery profile and an expanded deployed profile. FIG. 5a illustrates an elongated delivery profile where the central catheter (**60**) extends distally from within the axial lumen (**76**) of the bow catheter (**70**). The bow wires (**30, 40**) are in a generally straightened shape. The distal end (**62**) of the central catheter (**60**) and the distal end (**72**) of the bow catheter (**70**) are far apart from each other to the length of the bow wires (**30, 40**). And the two delivery catheter for tissue piercing wires (**110, 120**) are packed closely to the central catheter (**60**) and to bow catheter (**70**) to form a small radial profile.

(54) FIG. 5b further illustrates a deployment of the delivery catheter for tissue piercing wire (**110, 120**). In this deployed profile, the central catheter (**60**) retracts proximally, drawing its distal end (**62**) to the distal end (**72**) of the bow catheter (**70**) and the bow wires (**30, 40**) are outwardly radially away from the longitudinal axis of the bow catheter (**70**). The portion of the central catheter (**60**) outside of the bow catheter (**70**) and the radially curved bow wires form a generally flat surface. Following the radial movement of the bow wires (**30, 40**), the distal portions (**118, 128**) of the delivery catheter for tissue piercing wires (**110, 120**) are directed by the bow wires (**30, 40**) and pivot outwardly radially from the elongated axis of the central catheter (**60**) and the bow catheter (**70**).

(55) Similar to described above, especially in relation to FIGS. 3a-3b, the mitral commissure locating system (**50**) along with the delivery catheter for tissue piercing wire (**110, 120**) are delivered and deployed across the mitral valve (**6**). Tracking along a guide wire (**80**) extending across the mitral valve, the mitral commissure locating system (**50**) is first delivered to the mitral valve (**6**), with its distal portion extending through the slit between the posterior (**10**) and anterior mitral leaflets (**8**). Then, as a clinician draws the distal ends (**62, 72**) of the central catheter (**60**) and the bow catheter (**70**) closer to each other, the bow wires (**30, 40**) curve outward. As the bow wires (**30, 40**) further arc outwardly, each wire makes contact with a mitral valve commissure (**12, 14**). As the distal ends (**62, 72**) of the central catheter (**60**) and the bow catheter (**70**) draw further closely by a clinician from outside of the body, the bow wires (**30, 40**) curve further outwardly until the portion of the bow wires (**30, 40**) contacting the mitral valve commissure (**12, 14**) are obstructed from expanding further radially and forming a narrow waist (**38, 48**) similar to what has been shown in FIG. 3b.

(56) In one embodiment where the distal portions (118, 128) of the delivery catheter for tissue piercing wire (110, 120) stationarily attaches to the bow wires (30, 40), the deployment of the bow wire (30, 40) directs the distal portion (118, 128) of the delivery catheter for tissue piercing wire (110, 120) radially outwardly. The distal ends (112, 122) of the delivery catheter for tissue piercing wire (110, 120) are positioned adjacent, approximate to, or against the mitral annulus (16) and inside the left ventricle (4) at or near the mitral valve commissure (12, 14). Alternatively, where the distal portion (118, 128) of the delivery catheter for tissue piercing wire (110, 120) slidably attaches to the bow wires (30, 40), after the bow wires (30, 40) mark the mitral valve commissure (12, 14), the delivery catheter for tissue piercing wires (110, 120) slides distally along the bow wires (30, 40) and the bow catheter (70) so that distal ends (112, 122) are positioned adjacent, approximate to, or against the mitral annulus (16) and inside the left ventricle (4) at or near the mitral valve commissure (12, 14),

(57) In another embodiment, as the bow wires (30, 40) are deployed, the distal ends (112, 122) of the delivery catheter for tissue piercing wires (110, 120) are positioned inside the left atrium (2). In such an event, the clinician can retract the entire mitral commissure locating system (50) along with the delivery catheter for tissue piercing wires (110, 120) proximally, so that the distal ends (112, 122) of the delivery catheter for tissue piercing wires (110, 120) are retracted back into the left ventricle (4) and positioned adjacent, approximate to, or against the mitral annulus (16).

Alternatively, instead of retracting the entire mitral commissure locating system (50) and the delivery catheter for tissue piercing wires (110, 120), the delivery catheter for tissue piercing wires (110, 120) slide proximally along the bow wires (30, 40) and bow catheter (70) so that the distal ends (112, 122) are positioned approximately to the mitral annulus (16).

(58) In another embodiment; as the bow wires (30, 40) are deployed, the distal ends (112, 122) of the delivery catheter for tissue piercing wires (110, 120) are at a position that is inside the left ventricle (4) and distant to the mitral annulus (6). In such an event, the clinician can advance the entire mitral commissure locating system (50) along with the delivery catheter for tissue piercing wires (110, 120) proximally, so that the distal ends (112, 122) of the delivery catheter for tissue piercing wires (110, 120) are advanced distally and positioned adjacent, approximate to, or against the mitral annulus (16). Alternatively, instead of advancing the entire mitral commissure locating system (50) and the delivery catheter for tissue piercing wires (110, 120), the delivery catheter for tissue piercing wires (110, 120) slide distally along the bow wires (30, 40) and the bow catheter (60) so that distal ends (112, 122) are positioned approximate to the mitral annulus (16).

(59) In yet another embodiment, the distal ends (112, 122) of the delivery catheter for tissue piercing wires (110, 120) include one or more radiopaque markers so that they can be easily visualized by using a radiographic imaging equipment such as with x-ray, magnetic resonance, ultrasound, or fluoroscopic techniques. With the visual assistance provided by the radiopaque marker, during the deployment of the mitral commissure locating system (50), a clinician can adjust the positioning of the system (50) and the delivery catheter for tissue piercing wire (110, 120) jointly or separately, so that the distal ends (112, 122) of the delivery catheter for tissue piercing wires (110, 120) remain approximate to the mitral annulus (16).

(60) Once the delivery catheter for tissue piercing wires are properly positioned, the tissue piercing wires are then advanced relatively to the delivery catheter for tissue piercing wire similar to what has been described above, including in FIG. 4b. Similarly, according to one embodiment, the distal ends of the tissue piercing wires have piercing tips which allow them to perforate the annulus. In another embodiment, one or both of the distal ends of tissue piercing wires has a radio frequency (RF) energy delivery tip to assist its crossing of the annulus. In one embodiment, the two tissue piercing wires are advanced simultaneously across the annulus. In another embodiment, the two tissue piercing wires are advanced sequentially across the annulus.

(61) According to some embodiments, the tissue piercing wires advance inside the wire delivery lumen (116, 126) of the delivery catheter for tissue piercing wire (110, 120) and cross the annulus.

According to some embodiments, the tissue piercing wires are preload inside the wire delivery lumen (116, 126) and the delivery catheter for tissue piercing wires (110, 120) having the tissue piercing wires disposed inside are then positioned at a treatment location. In some embodiments, after the delivery catheter for tissue piercing wires (110, 120) are positioned at the treatment location, the multiple tissue piercing wires then advance distally and cross the annulus. In certain embodiments, the multiple tissue piercing wires advance simultaneously. In certain embodiments, the multiple tissue piercing wires advance sequentially. In certain embodiments, the multiple tissue piercing wires advance in groups.

(62) According to some embodiments, the distal portion of the tissue piercing wire is configured to deflect or curl back to prevent inadvertent tissue damage. The ability to deflect or curl can be achieved by the geometrical construct of the tissue piercing wire, such as a flexible distal portion, by the physical property of the material used in making the wire, or by the shape memory property of the material used in making the tissue piercing wire. Those skilled in the art would be able to incorporate known techniques and/or materials to achieve this purpose without undue experimentation.

(63) With the tissue piercing wire crossed the mitral annulus (16) at the mitral valve commissure, the mitral commissure locating system (50) along with the delivery catheter for tissue piercing wire are then removed from the body, leaving the tissue piercing wire remain across the mitral annulus (16) and mark the mitral valve commissure. FIG. 6 illustrates two tissue piercing wires (130, 140) remaining across the mitral annulus (16) at the mitral valve commissures (12, 14), with the distal ends (132, 142) of the tissue piercing wires (130, 140) inside the left atrium and the proximal ends (not shown) of the tissue piercing wires (130, 140) extending percutaneously outside of the body. In one embodiment, the two tissue piercing wires (130, 140) are simultaneously or sequentially positioned through two delivery catheter for tissue piercing wires attached to a mitral valve commissure locating system (50), for example as explained in detail above. In one embodiment, the two tissue piercing wires (130, 140) are sequentially positioned through one delivery catheter for tissue piercing wire attached to a mitral valve commissure locating system (50). For example, upon deploying one tissue piercing wire across the mitral annulus (16) at the posteromedial commissure (12), as explained in greater detail above, a clinician transitions the mitral valve commissure locating system (50) from its expanded deployed profile back to its elongated delivery profile, and removes it proximally to detach from the first tissue piercing wire positioned across the mitral annulus. The clinician then torques the mitral valve commissure locating system (50) along with the delivery catheter fix tissue piercing wire, so that the distal end of the delivery catheter for tissue piercing wire is positioned near the anterolateral commissures (14). A second tissue piercing wire is then deployed across the mitral annulus (16) at the anterolateral commissures (14). According to one embodiment of the present teachings, the tissue piercing wires are placed at the commissures (14, 16).

(64) According to one embodiment of the present teachings, two tissue piercing wires are positioned across the mitral annulus. One skilled in the art should understand that any number of tissue piercing wires can be placed with the assistance of the mitral commissure locating system.

(65) Although the present teachings provide tissue piercing wires that are placed across the commissures, one skilled in the art should understand that tissue piercing wire can be placed not only at the commissure, but also at other locations along the annulus, for example, the P1, P2, or P3 area of the posterior annulus. For example, the path of the tissue piercing wire can be adjusted by either steering its delivery catheter or by adjusting the degree of bend of the bow wire. And subsequently, tissue anchors can be placed at the locations marked by the tissue piercing wire, and the annulus between the tissue anchors are then plicated as described below.

(66) According to one embodiment, the bow wires transition together from its generally straightened profile to its bend profile. In another embodiment, two bow wires can be adjusted independently of each other so that one wire can have a different degree of bend from the other

wire.

(67) According to one embodiment, the delivery catheter fix the tissue piercing wire attaches to the bow wires at its distal end. In another embodiment, the delivery catheter for the tissue piercing wire attaches to the bow catheter at the distal end of the bow catheter. In yet another embodiment, the delivery catheter for the tissue piercing wire can attach to either of the bow wires or the bow catheter at any location needed. Thus, the specific embodiments disclosed here should not be viewed as limiting.

(68) According to one embodiment, a first tissue anchor (**160**) can then be deployed over the tissue piercing wires (**130**, **140**) and across the mitral annulus (**16**) at the mitral valve commissures (**12**, **14**). According to some embodiments, as illustrated in FIG. 7, a first tissue anchor delivery catheter (**150**) tracks along the tissue piercing wire (**130**), across the mitral annulus (**16**), and into the left atrium (**2**). In certain embodiments, the first tissue anchor delivery catheter (**150**) is used to deliver a first tissue anchor (**60**) to the mitral annulus (**16**).

(69) While any first tissue anchoring devices known in the art can be used, the particular tissue anchor in the present teachings, as shown in FIG. 8, is collapsible. In various embodiments, the particular tissue anchor in the present teachings comprises a plurality of discrete, fiat, or flexible anchor elements coupled with a flexible tensile member. The anchor elements can be made from a surgical grade fabric material (e.g., a polyester material such as DACRON). In some instances, designed to promote tissue in-growth so that the anchors become at least in part encased in tissue over-time. The anchor elements are coupled to a tensile member, in this example, a suture, by threading the suture distally through the anchor elements and proximally through the anchor elements. A slip knot or another type of locking mechanism is formed so that when a proximal end portion of the tensile member is pulled, all of the anchor elements will be drawn together. This leaves a long “tail” of the suture leading from the anchor to the venous access site and the long “tail” can be used for the subsequent tensioning and plication, as described herein.

(70) Examples of tissue anchors and tissue anchor delivery catheters described in conjunction with the drawings of the present teachings have some similarities to those in U.S. patent application Ser. No. 12/273,670, filed on Nov. 19, 2008, entitled Tissue Anchor and Anchoring System, U.S. patent application Ser. No. 11/174,951, filed on Jul. 5, 2005, entitled First Tissue Anchor and Anchoring System and Methods of Using the Same, U.S. patent application Ser. No. 13/777,042, filed on Feb. 26, 2013, entitled Tissue Anchor and Anchoring System, each of which is incorporated by reference herein in its entirety. Although not shown in the exemplary figures, other suitable tissue anchoring devices can also be used. Examples of suitable tissue anchoring devices include, but are not limited to, tissue fasteners, tissue pledgets, or tissue staples etc.

(71) Still referring to FIG. 7, a first tissue anchor delivery catheter (**150**) holding a first tissue anchor (**160**) inside its longitudinal lumen (**152**) tracks along the wire (**130**), across the mitral annulus (**16**), and with the distal end (**152**) of the first tissue anchor delivery catheter (**150**) inside the left atrium (**2**). FIG. 9a illustrates that the first tissue anchor (**160**) is partially pushed distally outside of the distal end (**152**) of the first tissue anchor delivery catheter (**150**). Once the distal portion (**162**) of the first tissue anchor (**160**) or a sufficient amount of the anchor elements is exposed inside the left atrium (**2**) a clinician stops pushing the first tissue anchor (**160**) distally and retracts the first tissue anchor delivery catheter (**150**) proximally so that the distal end (**152**) of the first tissue anchor delivery catheter (**150**) moves proximally across the annulus (**16**) and back into the left ventricle (**4**). The clinician then exposes the proximal portion (**164**) of the first tissue anchor (**160**) or the remainder of the anchor elements of the first tissue anchor (**160**) within the left ventricle (**4**) by further retracting the first tissue anchor delivery catheter (**150**) proximally as shown in FIG. 9b.

(72) FIG. 9c illustrates the deployment of the first tissue anchor (**160**). A clinician pulls the proximal end of the tensile member (**166**) of the first tissue anchor (**160**) such that the anchor elements of the first tissue anchor (**160**) are drawn together against the opposite sides of the mitral

annulus (16), thereby securing the first tissue anchor (160) to the mitral annulus (16) at the first mitral valve commissure. As a result, as illustrated in FIG. 9c, the first tissue anchor (160) is deployed across the mitral annulus (16) at the posteromedial commissure (12) with the distal portion of the first tissue anchor (160) placed against the atrial side of the mitral annulus (16), the proximal portion of the first tissue anchor (160) placed against the ventricle side of the mitral annulus (16), and the tensile member (166) of the first tissue anchor (160) extending proximally through the lumen (156) of the first tissue anchor delivery catheter (150) to the outside of the body. One skilled in the art should understand that although FIG. 9c shows a first tissue anchor is deployed across the mitral annulus at the posteromedial commissure (12) the first tissue anchor can also be deployed across the mitral annulus at the anterolateral commissures (14). Thus, what has been disclosed here should not be viewed as limiting.

(73) In an alternative embodiment, after the first tissue anchor (160) is deployed across the mitral annulus, a clinician cinches/deploys the distal part of the tissue anchor on the atrial side of the annulus before retracts the anchor delivery catheter and deploys the proximal part of the anchor in the left ventricle. In a particular embodiment, a clinician exposes the distal portion of the tissue anchor inside the left atrium and cinches the anchor elements of the distal portion of the tissue anchor so that they are placed against the left atrial side of the annulus. In another embodiment, a clinician exposes and cinches the proximal portion of the tissue anchor from inside the left ventricle.

(74) With the first tissue anchor (160) securely deployed at the first mitral valve commissure across the mitral annulus (16), the clinician can deploy a second tissue anchor (170) at a second mitral valve commissure according to some embodiments of the present teachings. FIG. 10 illustrates an exemplary deployment of a second tissue anchor (170) across mitral annulus (16) at a second mitral valve commissure. According to some embodiments, similar to what is described in FIGS. 9a-9c, a clinician uses similar steps to deploy a second tissue anchor (170) through the second tissue piercing wire (140) positioned at the second mitral valve commissure. As illustrated in FIG. 10, the second tissue anchor (170) is deployed across the mitral annulus (16) at the anterolateral commissure (14) with the distal portion of the second tissue anchor (170) placed against the atrial side of the mitral annulus (16), the proximal portion of the second tissue anchor (170) placed against the ventricle side of the mitral annulus (16), and the tensile member (176) of the second tissue anchor (170) extending proximally through the lumen (186) of the second tissue anchor delivery catheter (180) to the outside of the body.

(75) According to some embodiments, a clinician can then apply tension to one or both of the tensile members (166, 176) of the first and second tissue anchors (160, 170). This tension pulls two tissue anchors (160, 170) closer to each other, thereby reducing the circumference of the mitral annulus. This tension, and the reduced distance between the two tissue anchors (160, 170), are maintained by directing a locker (184) along the tensile member (166, 176) towards the two tissue anchors (160, 170). The two wires (130, 140) and/or the two tissues anchor delivery catheters (150, 180) can then be retracted proximally and removed.

(76) Suitable lockers include those known in the art and those described in U.S. application Ser. No. 11/753,921 filed on May 25, 2007, entitled Lockers for Surgical Tensile Members and Methods of Using the Same to Secure Surgical Tensile Members, the disclosure of which is incorporated herein by reference. With the tensile members secured by the locker, the excess tensile members proximal to the locker can be removed by a cutter, for example, a cutter disclosed in U.S. patent application Ser. No. 11/935,054, filed on Nov. 5, 2007, entitled Suture Cutter and Method of Cutting Suture, the disclosure of which is incorporated herein by reference.

(77) FIGS. 11a-11d illustrate another embodiment of the mitral annulus (16) plication. As illustrates in FIG. 11a, according to one embodiment of the present teachings, upon the first tissue piercing wire (130) positioned across the mitral annulus (16) at the first mitral valve commissure, a bident catheter is introduced into the left ventricle (4) with one lumen of the bident catheter sliding

over the first tissue piercing wire (130). Examples of the bident lumen catheter are disclosed in U.S. patent application Ser. No. 13/282,139, filed on Oct. 26, 2011, entitled Hand Operated Device for Controlled Deployment of a First tissue anchor (160) and Method of Using the Same, the disclosure of which is incorporated herein by reference.

(78) A bident catheter, for example is a double lumen catheter with the distal portions of the two catheter members separating from each other and the rest portion of the two catheter members joined together. The distal portions of the two catheter members are kept close to each other during a percutaneous delivery. Once reaching the treatment location, the distal portion of the two catheter members are deployed to be radially apart from each other, either automatically by the construct of the bident catheter; or manually by a control mechanism from outside of the body. Each catheter lumen is used to deliver one tissue anchor, similar to what has been described above.

(79) Alternatively, a tissue anchor delivery catheter with a single lumen catheter and a translation mechanism allowing catheter to be controllably moved to a second location can be used instead of a bident catheter. Examples of the delivery catheter with a translation mechanism is disclosed in U.S. Patent Application Ser. No. 61/786,373, filed on Mar. 15, 2013, entitled Translation Catheters, Systems, and Method of Use Thereof the disclosure of which is incorporated herein by reference.

(80) In one embodiment of the present teachings, a bident tissue anchor delivery catheter (200) comprises a first catheter member (210) having a lumen (216) threaded over the first tissue piercing wire (130) at the first mitral valve commissure. Once the distal end (212) of the first catheter member (210) is positioned approximately to the mitral annulus (16) at the first mitral valve commissure, the distal portion of the second catheter member (220) is deployed radially and laterally away from the distal portion of the first catheter member (210). In one embodiment, the distal end (222) of the second catheter member (220) directs generally to the mitral annulus (16) at the P3 region generally along the posterior mitral annulus (16b), as illustrated in FIG. 11a. A third tissue piercing wire is then delivered, followed by a third tissue anchor (230) deployed across the mitral annulus (16) at the P3 region. As shown in FIG. 11b, two tissue anchors (160, 230) are deployed across the mitral annulus (16), one at the posteromedial commissure (12) and the other at the P3 region of the posterior annulus (16b). The distal portions of the two tissue anchors (160, 230) are placed against the atrial side of the mitral annulus (16), the proximal portions of the two tissue anchors (160, 230) are placed against the ventricle side of the mitral annulus (16), and the tensile member (166, 236) of the two tissue anchors (160, 230) extend proximally to the outside of the body.

(81) Similar to what has been described above, according to some embodiments, a clinician can then apply tension to one or both of the tensile members (166, 236) of the two tissue anchors (160, 230). This tension pulls two tissue anchors (160, 230) closer to each other, thereby plicating the annulus between the two tissue anchors (160, 230). This tension and the reduced distance between the two tissue anchors (160, 230) are maintained by directing a locker (238) along the tensile members (166, 236) towards the two tissue anchors (160, 230), as illustrated in FIG. 11c.

(82) According to some embodiments, similar to what is described in FIGS. 11a-11c, a clinician uses similar steps to introduce another bident catheter (or a catheter with a translation mechanism) into the left ventricle (4) with one catheter lumen of the bident catheter sliding over the second tissue piercing wire (140), and the distal end of the first catheter member positioned approximately to the mitral annulus (16) at the second mitral valve commissure and the distal end of the second catheter member directed generally to the mitral annulus (16) at the P1 region generally along the posterior mitral annulus. Two tissue anchors are then deployed across the mitral annulus (16) with one at the second mitral valve commissure and the other at the P1 region generally along the posterior mitral annulus. FIG. 11d illustrates a third and a fourth tissue anchors deployed across the mitral annulus with one at the anterolateral commissures (14) and the other one at the P1 region generally along the posterior mitral annulus. A clinician applies tension to one or both of the tensile members of the third and fourth tissue anchors and locks the tension with a locker, thereby

plicating the mitral annulus (16) between the two anchors, as shown in FIG. 11d.

(83) FIGS. 12a-12b illustrate four tissue anchors deployed across the mitral annulus (16) along the posterior annulus according to one embodiment of the present teachings. The first tissue anchor is deployed at the posteromedial commissure (12), the second one is deployed at the P3 region generally along the posterior mitral annulus, the third one is deployed at the anterolateral commissure (14), and the fourth one is deployed at the P1 region generally along the posterior mitral annulus. In one embodiment, a clinician applies tension to at least one of the four tissue anchors and locks such tension with one locker, as illustrated in FIG. 12a. In another embodiment, a clinician locks the tension between the adjacent tissue anchors with one locker as illustrated in FIG. 12b.

(84) Without limiting the present teachings to any particular theory, the commissure areas have generally better structures than other areas since they are closer to the fibrous skeleton of the heart and, therefore may provide a stronger base to pull the dilated posterior annulus in the antero-posterior direction. Mitral annulus plication with one tissue anchor placed at a commissure provides a significant dimensional change in the antero-posterior direction which, without limiting the present teachings to any particular theory, is generally thought to be important for correcting a mitral regurgitation caused by a dilated annulus. In various embodiments of the present teachings the mitral commissure locating system uses the anatomical geometry of a heart for precisely placing tissue anchors described in the present application. Using the mitral commissure locating system allows positioning of the tissue anchor in a position similar to the Kaye Annuloplasty surgical procedure.

(85) The mitral commissure locating system (50) disclosed herein is useful for delivering positioning wires and tissue anchors across the mitral annulus (16) at the mitral commissure and plicating the mitral annulus without potentially being caught by the tendinous chords during the process. One skilled in the art will further recognize that the present teachings could be used to reshape the tricuspid annulus or other heart valve annulus.

(86) Although the present teachings disclose deployments of tissue anchors over a wire placed across the mitral annulus, one skilled in the art should understand that tissue anchors can also be delivered with a tissue anchor delivery catheter tracking along the bow wires of the mitral valve commissure locating system as disclosed above. Thus, the specific step of the tissue anchor deployment is subject to change as needed by a person with ordinary skill in the field. Specific embodiments disclosed in the present teachings should not be viewed as limiting.

(87) Various embodiments have been illustrated and described herein by way of examples, and one of ordinary skill in the art will appreciate that variations can be made without departing from the spirit and scope of the present teachings. The present teachings are capable of other embodiments or of being practiced or carried out in various other ways, for example in combinations. Also, it is to be understood that the phraseology and terminology employed herein is for the purpose of description and should not be regarded as limiting.

(88) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this present teachings belong. Methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present teachings. In case of conflict, the patent specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.

Claims

1. A method, comprising: transluminally advancing a system into a subject; positioning a first portion of the system adjacent a cardiovascular tissue of the subject, the system including wires disposed at the first portion; stabilizing the first portion at the cardiovascular tissue by positioning,

against a cardiovascular wall of the subject, a distal portion of the system, the distal portion being distal to the first portion; while the distal portion of the system remains against the cardiovascular wall, inducing the wires to bow laterally such that the wires press against the cardiovascular tissue; while the wires remain pressed against the cardiovascular tissue, radiographically imaging a shape of the wires; and while (i) the first portion remains stabilized, and (ii) the wires remain pressed against the cardiovascular tissue, transluminally implanting a second portion of the system at the cardiovascular tissue, facilitated by the imaging.

2. The method according to claim 1, wherein, for each of the wires, a first segment of the wire is disposed at the first portion and a second segment of the wire is disposed at the distal portion, and wherein inducing the wires to bow laterally comprises inducing the wires to bow laterally such that the second segment of each of the wires moves laterally and presses against the cardiovascular tissue.

3. The method according to claim 1, wherein: the wires are first wires, the system includes second wires disposed at the distal portion, and positioning the distal portion against the cardiovascular wall comprises positioning the second wires within a blood vessel of the subject.

4. The method according to claim 3, wherein the method further includes securing the distal portion inside the blood vessel by inducing the second wires to bow laterally within the blood vessel.

5. The method according to claim 1, wherein inducing the wires to bow laterally such that the wires press against the cardiovascular tissue comprises inducing the wires to bow laterally such that the wires press against a cardiac valve.

6. The method according to claim 5, wherein the system includes exactly two wires, and wherein inducing the wires to bow laterally such that the wires press against the cardiovascular tissue comprises inducing the exactly two wires to bow laterally such that the exactly two wires press against the cardiac valve.

7. The method according to claim 1, wherein: the system includes a bow catheter, and a central catheter that is slidably disposed within a lumen of the bow catheter, each of the wires has a first end attached to the bow catheter, and a second end attached to the central catheter, and inducing the wires to bow laterally comprises inducing the wires to bow laterally by sliding the central catheter with respect to the bow catheter.

8. The method according to claim 7, wherein positioning the first portion of the system at the cardiovascular tissue comprises positioning the first portion of the system such that the first end of each of the wires is disposed downstream of a cardiac valve, and the second end of each of the wires is disposed upstream of the cardiac valve.

9. The method according to claim 7, wherein inducing the wires to bow laterally by sliding the central catheter with respect to the bow catheter comprises inducing the wires to bow laterally by retracting the central catheter into the bow catheter.

10. The method according to claim 1, further comprising transluminally advancing a delivery catheter to the cardiovascular tissue, and delivering the second portion of the system to the cardiovascular tissue via the delivery catheter.

11. The method according to claim 10, further comprising tracking the delivery catheter along one of the wires.

12. The method according to claim 1, wherein implanting the second portion of the system comprises implanting at least a portion of an implant at a commissure of a cardiac valve.

13. The method according to claim 1, wherein the second portion of the system is a first implant, and wherein the method further comprises transluminally anchoring a second implant to cardiovascular tissue, facilitated by the imaging.

14. The method according to claim 13, wherein implanting the first implant comprises implanting the first implant at a first commissure of a cardiac valve, and wherein implanting the second implant comprises implanting the second implant at a second commissure of the cardiac valve.

15. The method according to claim 13, further comprising, subsequently to implanting the first and

second implants, transluminally reducing a distance between the first and second implants by applying tension to a tensile member that is coupled to the first and second implants.

16. The method according to claim 15, further comprising locking the tension in the tensile member by locking a locker to the tensile member.

17. The method according to claim 16, further comprising, prior to locking the locker to the tensile member, advancing the locker along the tensile member toward the first and second implants.

18. The method according to claim 16, further comprising, subsequently to locking the locker to the tensile member, using a cutter to remove excess of the tensile member.

19. The method according to claim 1, wherein the system includes a catheter, the wires disposed laterally from the catheter, and wherein positioning the first portion of the system adjacent the cardiovascular tissue comprises positioning the system such that the catheter extends beyond the cardiovascular tissue.

20. The method according to claim 1, wherein inducing the wires to bow laterally such that the wires press against the cardiovascular tissue comprises inducing the wires to bow laterally until the wires form a waist at the cardiovascular tissue.

21. The method according to claim 1, wherein the cardiovascular wall is a wall of a vein of the subject, and wherein positioning the distal portion against the cardiovascular wall comprises positioning the distal portion against the wall of the vein.
